SmarTest Pro / SnapTest Pro — System Map

Build 240 | web 1.4.26 | Firebase project: the-vishnu-sharma-test (Spark / free plan)

A. USER APPS (sabka code ek hi repo me)

Student App

Web (PWA) + Android APK
Login: mobile + PIN
Tests dena, result, admit card,
daily challenge, notifications

Admin / Teacher App

Wahi Web + APK, role login se
Login: email + password
Tests, question bank, records,
batches, OMR, fees, students

Owner App

owner-app.html + Owner APK
com.snaptestpro.owner
Tabs: Home, Institutes, Alerts,
Reads (naya), Settings

Institute APK

Har institute ka apna APK
Owner Panel -> 'APK banao'
Branding + lock (appLock)
GitHub Actions banata hai

app ka code + APK download (sab apps ko)

SDK: Auth + Firestore (direct, rules + App Check)

HTTPS: /bank, /student/*, /readlog

B. DELIVERY, BACKEND AUR DATA

GitHub (code + hosting)

Repo: SnaptestPro/snaptestpro.github.io

GitHub Pages

Site: snaptestpro.github.io

Student/Admin/Owner web yahin se

sw.js = offline cache

GitHub Releases

APK files + app-version.json

GitHub Actions (workflows)

auto-extract: zip upload -> files

build-android / institute-apk /

owner-apk, ci, announce-update

Secrets: signing, WORKFLOW_PAT

Firebase / Google Cloud project

Authentication

Admin + Owner: email/password

Student: anonymous (+ PIN check)

(OFF) custom-token login (serverAuth)

Cloud Firestore (main database)

institutes/{id}/ batches, subjects,

chapters, items, roster, ver

tests (+ qchunks), students,

studentRecords, admins, questionBank,

questionReports, deletedQuestions

Rules = access control

App Check

APK: config ON | Web: key khali (OFF)

Cloud Messaging (FCM)

Push notification to phones

Storage

(band) SDK load hota hai; upload Drive se

Quota (Spark): 50K reads/din, 20K writes/din

REST + service account

Cloudflare Worker

cool-thunder-a280.workers.dev

/bank, /bank/chunk question bank

/student/login, /register, /set-pin,

/reset-with-pin, /delete-account

/alert, /alerts, /broadcast push+alert

/readlog + /owner/reads/* Reads Monitor

/owner/apk, /owner/recount

Cron: reminders, nightly recount,

daily expiry digest

Secrets: service account, owner emails,

Telegram, GitHub token

CDN (libraries, free)

cdnjs.cloudflare.com, cdn.jsdelivr.net,
unpkg.com -> JS libraries
gstatic.com -> Firebase SDK 9.22 (compat)
Google Fonts -> fonts

Worker ke andar ka data

KV (BANK_KV): bank chunks, alerts history

D1 (READS_DB): Reads Monitor logs

(band) R2 / FILES_KV: file store band hai

C. GOOGLE AUR BAAKI SERVICES

apps se: Drive upload, YouTube, Forms

Worker se: Telegram, GitHub API

Google Cloud Console + Google APIs

IAM service account: Worker Firestore/FCM ke liye
OAuth Client ID: Google Drive login (drive upload)
Google Drive: institute ki files (PDF/photo/video)
(OFF) Cloud Monitoring API: Billing (Blaze) chahiye
-> Reads tab me 'haath se mel' box

Google Forms / Apps Script / YouTube

Exam registration: auto Google Form + Apps Script
web-app (script.google.com) se responses pull
YouTube: live class (embed)
Google Docs viewer: PPT/DOC/XLS dikhane ke liye

Baaki (outside)

Telegram bot: Worker se owner alert (optional)
WhatsApp (wa.me links): parent report, support
ChatGPT: sirf copy-paste (OMR prompt), API nahi

Chhote kaam: FCM push -> Firebase se seedha phone par. Laal text (OFF)/(band) = config me band ya use nahi ho raha. Ye chart code se banaya gaya hai (v240).